ACT SUBJECT TEST & DIGITAL SAT MATH COURSES

Rabaa Farrag

01120486569

23 YEARS OF EXPERIENCE

Reach a 700+ in ACT, Digital SAT & EST Math

✓ FIRST SESSION IS FREE!

Don't watch others hit 700+ while you wait.

ACT2 • ACT SUBJECT TEST MATH LEVEL 1 • PRACTICE EXAM

ACT2 Exam 4

Full Practice Test • Test #4

Area, slopes, factoring, similarity, statistics & circles. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	60 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

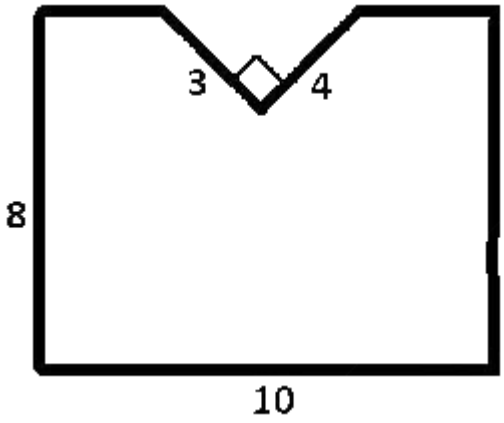
ACT, Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

ACT SUBJECT TEST • MATH LEVEL 1 • TEST #4

Practice Questions

50 QUESTIONS

1 Marvin removed a triangle section from a rectangle, as shown below. Lengths in the figure are given in meters. What is the area, in square meters, of the remaining polygon?



(A) 68
(B) 74
(C) 75
(D) 78

2 A line passes through the points $(4, 1)$ and $(7, 10)$. What is the slope of the line?

(A) -3
(B) $-\frac{1}{3}$
(C) $\frac{1}{3}$
(D) 3

3 Which of the following shapes has the greatest number of lines of symmetry?

(A) Square
(B) Rectangle that is not a square
(C) Equilateral triangle
(D) Circle

4 A circular pizza fits tightly in the bottom of a square box. The inside area of the box is 144 square inches. Which of the following values is closest to the area, in square inches, of the top of the pizza?

(A) 144π
(B) 36π
(C) 12π
(D) 6π

Rabaa Farrag

5 Given points A, B, C, D, and E are collinear. $AC = 12$, $DB = 20$, $BE = 8$. C is between B and E, A is between B and D, and B is the midpoint of $\overline{AC}$. What is DE?

(A) 40
(B) 28
(C) 24
(D) 16

6 The value of a car as it ages is displayed in the table below. Based on the table, which of the following completes the sentence: As the age of the car increases, the value of the car decreases ____.

Age (Yr)	Value
0	\$25,000.00
3	\$15,353.13
5	\$11,092.63
9	\$5,790.42
12	\$3,556.04

(A) Linearly by \$9,646.87 per year
(B) Linearly by \$1,787.00 per year
(C) Exponentially by 85% per year
(D) Exponentially by 15% per year

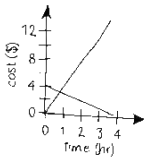
7 Which of the following equations represents the line in the standard (x, y) coordinate plane that has y-intercept of -2 and that is parallel to the line represented by the equation $y = \frac{2}{3}x - 4$?

(A) $y = -\frac{2}{3}x - 2$
(B) $y = -\frac{3}{2}x - 2$
(C) $y = \frac{2}{3}x - \frac{2}{3}$
(D) $y = \frac{2}{3}x - 2$

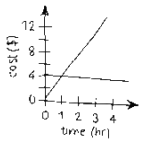
Rabaa Farrag

8 Jayme and her friends are planning a bowling party. The family bowling center offers 2 different party rate plans. Plan A has a cost of $4.00 per person per hour with no charge for shoe rental. Plan B has a cost of $2.00 per person per hour and $4.00 per person for shoe rental. Jayme graphs the cost per person for each plan so that she and her friends can choose the less expensive plan for the amount of time they want to bowl. Jayme creates one of the following graphs. Which one?

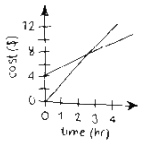
A.



B.



C.



(A) Graph A
(B) Graph B
(C) Graph C

9 Which of the following expressions is a factor of $9x^2 - 48x + 64$?

(A) $3x - 32$
(B) $3x - 8$
(C) $9x - 4$
(D) $9x - 16$

10 A line passes through quadrant 1 and quadrant 3 in the standard (x, y) coordinate plane. Which statement about the slope of this line is true?

(A) It is positive
(B) It is negative
(C) It is zero
(D) It is undefined

Rabaa Farrag

11 One of the following tables shows a direct variation between x and y. Which one?

A.

X	Y
-2	-5
1	-2
4	1

B.

X	Y
-2	4
1	4
4	4

C.

X	Y
-2	6
1	-3
4	-12

D.

X	Y
-2	14
1	8
4	2

(A) Table A
(B) Table B
(C) Table C
(D) Table D

12 Simplify $6x + 8 \div 2 - (4x - 5)$.

(A) $2x + 9$
(B) $2x - 1$
(C) $-x + 9$
(D) $-x - 1$

13 Simplify $\dfrac{7(4-3) - 6\left(\frac{1}{2} - \frac{1}{3}\right)}{6} + 7(m-1)$

(A) $7m$
(B) $7m - 6$
(C) $7m + 8$
(D) $\frac{7}{6}m - 1$

14 If isosceles triangle $\triangle ABC$ has an obtuse angle at vertex B, which statement is true?

(A) $AB + BC < AC$
(B) $AB = AC$
(C) $AB = BC$
(D) $AC = BC$

15 Triangle $\triangle ABC$ is a right isosceles triangle with $AB > AC$. Find $m\angle B$.

(A) $30°$
(B) $45°$
(C) $60°$
(D) $90°$

Rabaa Farrag

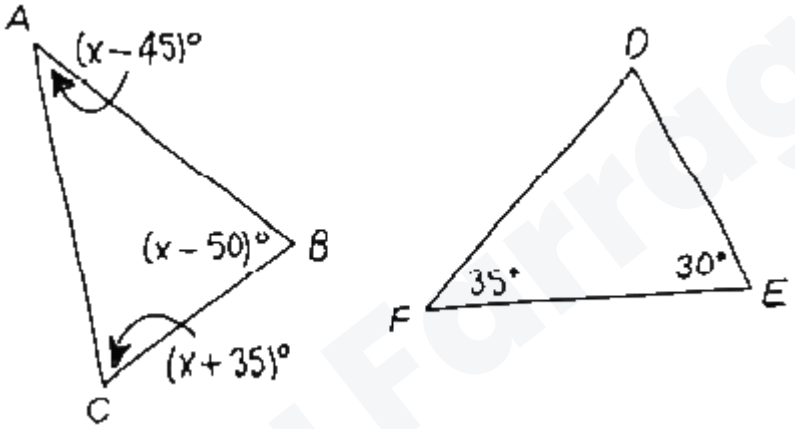
16 The radius and height of a cylinder are in the ratio $5 : 7$, and its volume is $550\ \text{in}^3$. What is the radius of the cylinder, in inches?

(A) $\sqrt{5}$
(B) $\sqrt{7}$
(C) 5
(D) 7

17 What values of y make this equation true? $|2y + 1| = 7$

(A) -3 and 4
(B) -3 and 3
(C) -4 and 4
(D) -4 and 3

18 If $\triangle ABC \cong \triangle FED$, with angle measures as noted, which equation CANNOT be used to find the value of x?



(A) $x + 35 = 115$
(B) $x - 50 = 30$
(C) $x - 45 = 35$
(D) $x + 35 = 65$

19 A rectangle has adjacent sides measuring $(x + 3)$ inches and $2x$ inches. The perimeter of the rectangle is 51 inches. Determine the area of the rectangle to the nearest 0.1 square inch.

(A) 120.3
(B) 157.5
(C) 225.0
(D) 608.0

20 Simplify the expression $\sqrt{6}\left(5 + \sqrt{6}\right)$.

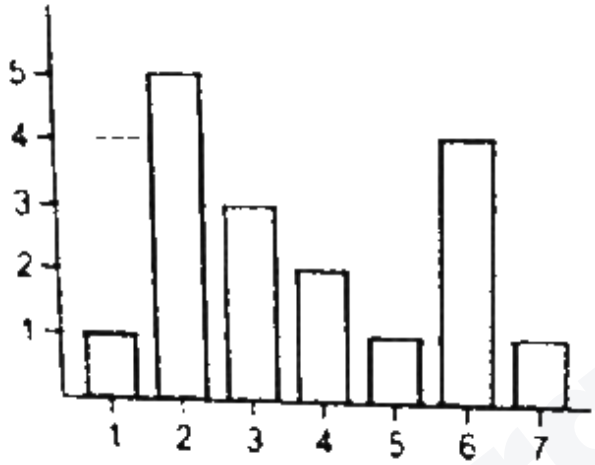
(A) $\sqrt{30} + 6$
(B) $6\sqrt{5} + 6$
(C) $5\sqrt{6} + 6$
(D) $11\sqrt{6}$

Rabaa Farrag

21 After taking 3 exams in physics, Zola's mean score is 80. If she scores 70 on her 4th exam, what will be her new mean score?

(A) 77.5
(B) 75
(C) 72.5
(D) 65

22 Based on this bar graph, what is the median of this set of data?



(A) 3
(B) 3.5
(C) 4
(D) 5

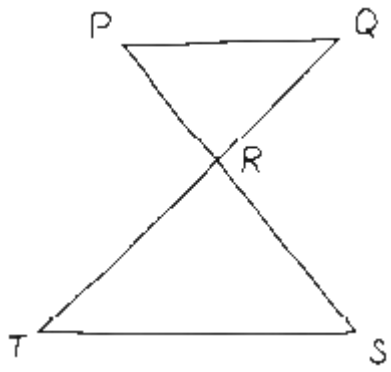
23 This chart shows the number of people who visited a state park during the years shown. Assuming that this geometric growth trend continued, which of the following values best estimates the number of people who visited the park in the year 2000?

Year	Thousands of individuals
1970	3
1975	6
1980	12
1990	48
2000	?

(A) $60{,}000$
(B) $69{,}000$
(C) $96{,}000$
(D) $192{,}000$

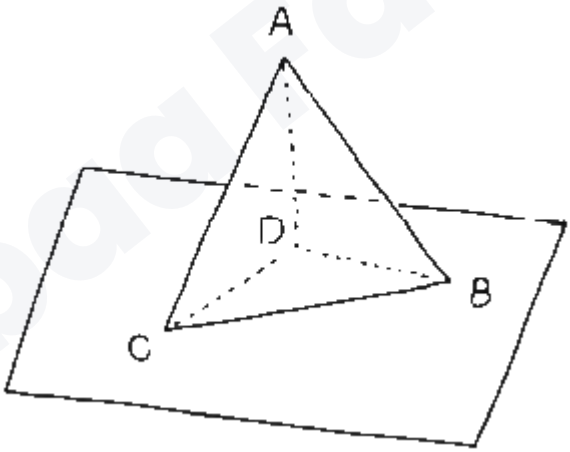
Rabaa Farrag

24 The figure below shows $\triangle PQR$ and $\triangle RST$ where R is the intersection of $\overline{PS}$ and $\overline{TQ}$. Which of the following statements, if true, could be used to prove $\triangle PQR \sim \triangle SRT$ without any other given information?



A $\overline{PQ} \parallel \overline{TS}$
B $\overline{PQ} \cong \overline{TS}$
C $\overline{PR} \cong \overline{TR}$
D $\overline{PS} \perp \overline{TS}$

25 In this figure: $\overline{AD}$ is perpendicular to plane BCD, $\overline{CD} \perp \overline{DB}$, $CD = DB$, and $m\angle DAB = 35°$. What is $m\angle ACD$?



A $35°$
B $45°$
C $55°$
D $90°$

26 Solve for x where a, b, and c are real numbers and $a > 0$: $ax + b > 3b + c$

A $x > \frac{c+2b}{a}$
B $x < \frac{c+2b}{a}$
C $x > c + 2b - a$
D $x < \frac{c}{a} + 2b$

Rabaa Farrag

27 Given that $f(2) = -20$, what are the zeros of $f(x) = x^2 + 3x + bx + 3b$?

(A) -6 and 3
(B) -3 and 2
(C) -3 and 6
(D) -2 and 3

28 For all positive values of x and y, the expression $(3x^0y^2)(2x^2y^3)^2$ is equivalent to:

(A) $6x^2y^5$
(B) $12x^4y^7$
(C) $12x^4y^8$
(D) $36x^4y^{10}$

29 Kristi has a photo that is 4 inches tall and 6 inches long. She enlarges the photo such that the ratio of the dimensions remains the same and the new photo is 5 inches longer than it is tall. What is the new photo's length, in inches?

(A) 9
(B) 10
(C) 15
(D) 20

30 Given $x = 3$ and $y = -2$, what is the value of $\frac{4x^5y^2}{15x^2y^3}$?

(A) $-\frac{18}{5}$
(B) $-\frac{6}{5}$
(C) $\frac{12}{5}$
(D) $\frac{18}{5}$

31 A plane flew 662 miles from Cleveland to New Orleans in 2 hours 20 minutes. To the nearest mile per hour, what was the average speed of the plane?

(A) 221
(B) 284
(C) 301
(D) 331

32 What are the zeros of the function $f(x) = x^2 - 3x - 4$?

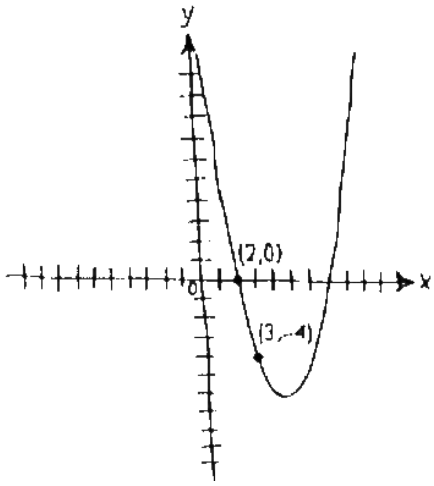
(A) -1 and 4
(B) -4 and -3
(C) -4 and 1
(D) $-4, -1, 1$, and 4

33 Rationalize the denominator of $\frac{2}{\sqrt{3}}$.

(A) $2\sqrt{3}$
(B) $-2\sqrt{3}$
(C) $\frac{2\sqrt{3}}{\sqrt{6}}$
(D) $\frac{2\sqrt{3}}{3}$

Rabaa Farrag

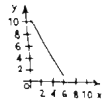
34 What equation corresponds to this graph?

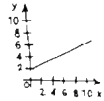


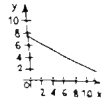
(A) $y = -4x + 8$
(B) $y = x^2 - 5x + 6$
(C) $y = x^2 - 9x + 14$
(D) $y = x^3 - 11x^2 + 32x - 28$

35 Let a and b be constants such that one of the following graphs in the standard (x, y) coordinate plane contains all the points in the table below. Which one?

A.


B.


C.


D.


x	y
a	b
$a+1$	$b-2$
$a+2$	$b-4$
$a+3$	$b-6$

(A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D

Rabaa Farrag

36 For all values of $y > 0$, which expression is equivalent to $\frac{x}{\sqrt{y}}$?

A x
B $x\sqrt{y}$
C $\frac{\sqrt{xy}}{y}$
D $\frac{x\sqrt{y}}{y}$

37 A science student observed that 9 times the Celsius temperature (C) of an object is equal to 160 less than 5 times the Fahrenheit temperature (F) of that object. Which of the following equations models the physical phenomenon that the student observed?

A $F = \frac{9}{5}C + 32$
B $F = \frac{5}{9}C - 32$
C $F = \frac{5}{9}C + 32$
D $F = \frac{9}{5}C - 32$

38 Each wheel on Shing's bicycle has a 14 inch radius. How many complete rotations will each wheel make in traveling a distance of 1 mile? (Note: 1 mile $= 5{,}280$ ft and $\pi \approx 3.14$)

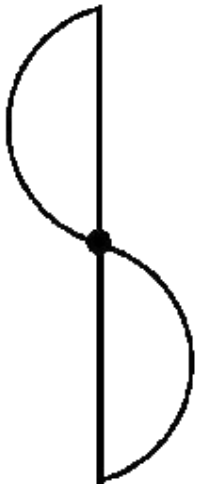
A 60
B 100
C 120
D 720

39 Which of the following sets gives both solutions to the quadratic equation $4x^2 - 16x + 15 = 0$?

A $\{-\frac{3}{2}, \frac{5}{2}\}$
B $\{\frac{3}{2}, -\frac{5}{2}\}$
C $\{-\frac{3}{2}, -\frac{5}{2}\}$
D $\{\frac{3}{2}, \frac{5}{2}\}$

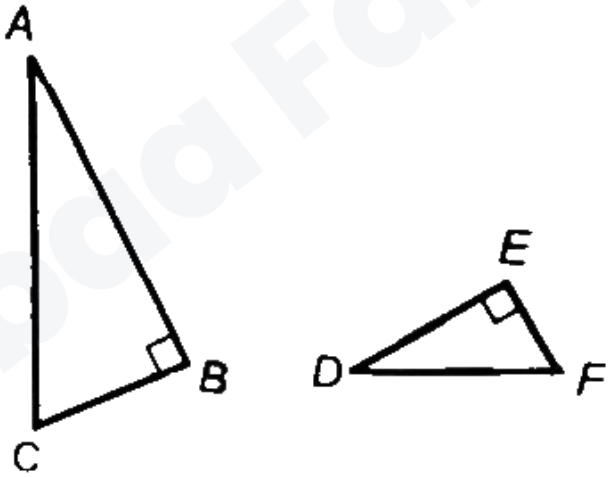
Rabaa Farrag

40 The figure below consists of 2 semicircles and 2 diameters that intersect in 1 point. The diameters are congruent and collinear. How many lines of symmetry and how many points of symmetry does this figure have, if any?



A 0 lines and 0 points
B 0 lines and 1 point
C 1 line and 0 points
D 1 line and 1 point

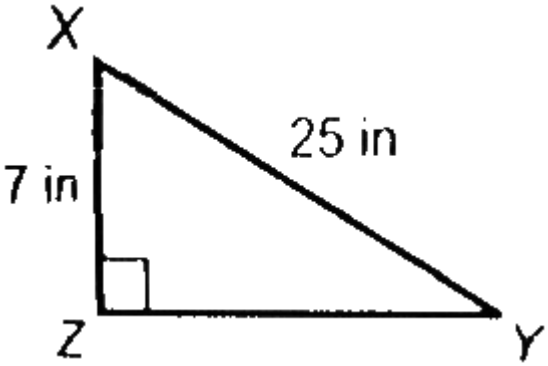
41 In right triangles $\triangle ABC$ and $\triangle DEF$ shown below, $\angle C \cong \angle F$, $BC = 5$ cm, $AB = 8$ cm, and $DE = 4$ cm. What is the length of $\overline{EF}$ in cm? (Note: the triangles are not drawn to scale.)



A $\frac{5}{32}$
B $\frac{5}{2}$
C 1
D $\frac{32}{5}$

Rabaa Farrag

42 In right triangle $\triangle XYZ$ shown below, what is ZY, in inches?



A $7\sqrt{3}$
B $\sqrt{674}$
C 18
D 24

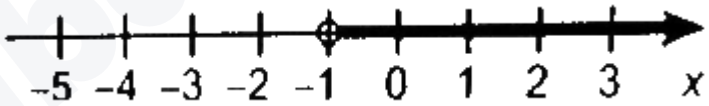
43 Simplify $2\sqrt{2} \cdot 3\sqrt{5}$.

A $5\sqrt{7}$
B $5\sqrt{10}$
C $6\sqrt{7}$
D $6\sqrt{10}$

44 Which of the following expressions is equivalent to $4x^3y^2\left(2y^4x + 6x^3y^7\right)$?

A $8x^3y^8 + 24x^9y^{14}$
B $8x^{12}y^2 + 24x^9y^{14}$
C $8x^4y^6 + 24x^6y^9$
D $8x^7y^3 + 24x^6y^9$

45 Which of the following inequalities is represented on the number line below?



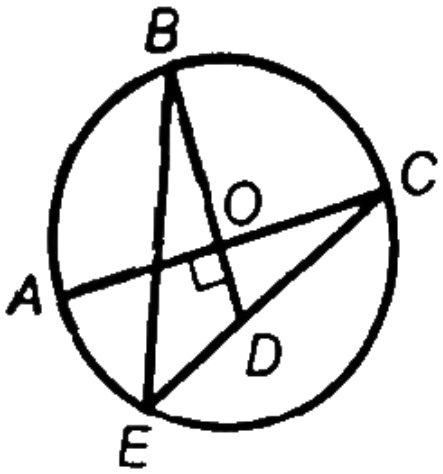
A $x \geq -1$
B $x \leq -1$
C $x > -1$
D $x < -1$

46 What is the name of a polygon in which the sum of the interior angles is $540°$?

A Pentagon
B Hexagon
C Heptagon
D Octagon

Rabaa Farrag

47 Given points A, B, C, and E lie on circle O. $\overline{AC} \perp \overline{BD}$ at O. $\overline{AC}$ is a diameter of the circle. E, D, and C are collinear. What is $m\angle BEC$?

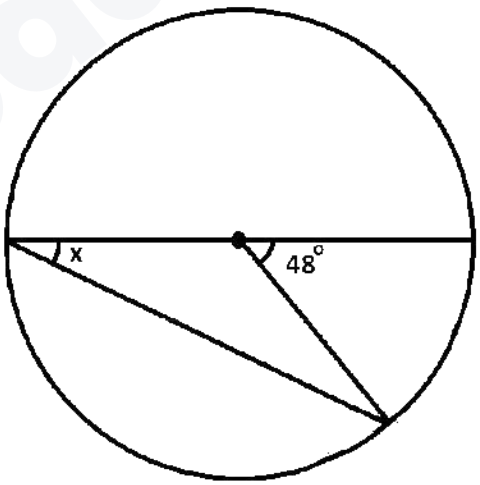


A) $22.5°$
B) $30°$
C) $45°$
D) $60°$

48 The base of a right triangular incline is 67.2 yd. The distance covered when walking up the incline is 84 yd. What is the height, in yards, of the incline?

A) 107.6
B) 50.4
C) 33.6
D) 16.8

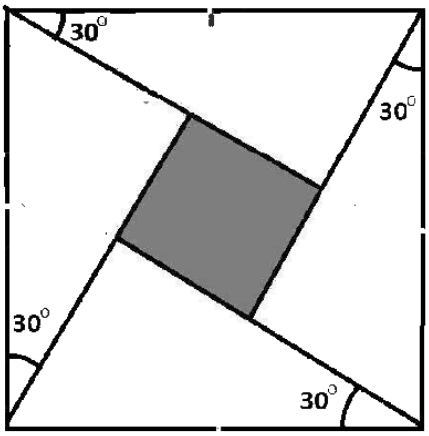
49 Find the value of x.



A) 96
B) 24
C) 52
D) 48

Rabaa Farrag

50 Find the area of the shaded region. (Round your answer to the nearest 0.1)



A) 70.18
B) 77.17
C) 65.8
D) 75.6

Rabaa Farrag

TEST #4 • CHECK YOUR WORK

Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	B	26	A
2	D	27	C
3	D	28	C
4	B	29	C
5	B	30	A
6	D	31	B
7	D	32	A
8	C	33	D
9	B	34	C
10	A	35	B
11	C	36	D
12	A	37	A
13	B	38	D
14	C	39	D
15	B	40	B
16	C	41	B
17	D	42	D
18	D	43	D
19	B	44	C
20	C	45	C
21	A	46	A
22	A	47	C
23	D	48	B
24	A	49	B
25	C	50	B

Rabaa Farrag

TEST #4 • STEP-BY-STEP

Detailed Solutions

50 SOLUTIONS

1 **Answer: B** 74

Step 1: Area of the full rectangle: $10 \times 8 = 80\ \text{m}^2$.
Step 2: The removed piece is a right triangle with legs 3 and 4: area $= \frac{1}{2}(3)(4) = 6\ \text{m}^2$.
Step 3: Remaining area $= 80 - 6 = 74\ \text{m}^2$.

2 **Answer: D** 3

Step 1: Slope formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$.
Step 2: $m = \frac{10-1}{7-4} = \frac{9}{3} = 3$.

3 **Answer: D** Circle

Step 1: Square: 4 lines; rectangle (not square): 2; equilateral triangle: 3.
Step 2: A circle is symmetric about every diameter — infinitely many lines, the greatest.

4 **Answer: B** 36π

Step 1: The box is a square of area 144, so its side is $\sqrt{144} = 12$ in.
Step 2: A circle fitting tightly has diameter 12, so $r = 6$.
Step 3: Area $= \pi(6)^2 = 36\pi\ \text{in}^2$.

5 **Answer: B** 28

Step 1: B is the midpoint of $\overline{AC}$ and $AC = 12$, so $AB = BC = 6$.
Step 2: A is between B and D and C is between B and E, so D and E lie on opposite sides of B: order D, A, B, C, E.
Step 3: $DE = DB + BE = 20 + 8 = 28$.

6 **Answer: D** Exponentially by 15% per year

Step 1: Check for a constant difference: $25000 \to 15353$ in 3 years is not a constant drop per year (later drops are smaller), so it is not linear.
Step 2: Check for a constant ratio per year: $\frac{15353.13}{25000} = 0.6141 = 0.85^3$, $\frac{11092.63}{15353.13} = 0.7225 = 0.85^2$.
Step 3: Each year the value is multiplied by 0.85: it decreases exponentially by 15% per year.

Rabaa Farrag

7 **Answer: D** $y = \frac{2}{3}x - 2$

Step 1: Parallel lines have equal slopes, so the new slope is $\frac{2}{3}$.
Step 2: The y-intercept is -2.
Step 3: $y = \frac{2}{3}x - 2$.

8 **Answer: C** Graph C

Step 1: Plan A: $C = 4t$ — starts at 0 and rises by 4 per hour.
Step 2: Plan B: $C = 2t + 4$ — starts at 4 and rises by 2 per hour.
Step 3: They are equal when $4t = 2t + 4 \Rightarrow t = 2$ (cost $\$8$). Graph C shows one line from 0, one from 4, crossing at $(2, 8)$.

9 **Answer: B** $3x - 8$

Step 1: $9x^2 = (3x)^2$, $64 = 8^2$ and $48x = 2(3x)(8)$.
Step 2: So $9x^2 - 48x + 64 = (3x - 8)^2$, and $3x - 8$ is a factor.

10 **Answer: A** It is positive

Step 1: Quadrant I has $(+, +)$ points and quadrant III has $(-, -)$ points.
Step 2: A line through both rises from lower-left to upper-right, so its slope is positive.

11 **Answer: C** Table C

Step 1: Direct variation means $y = kx$, so $\frac{y}{x}$ is the same for every pair (and the line passes through the origin).
Step 2: Table C: $\frac{6}{-2} = -3$, $\frac{-3}{1} = -3$, $\frac{-12}{4} = -3$ — constant.
Step 3: The other tables do not have a constant ratio.

12 **Answer: A** $2x + 9$

Step 1: Order of operations: division first, $8 \div 2 = 4$.
Step 2: $6x + 4 - (4x - 5) = 6x + 4 - 4x + 5$.
Step 3: $= 2x + 9$.

13 **Answer: B** $7m - 6$

Step 1: Numerator: $7(4 - 3) = 7$ and $6\left(\frac{1}{2} - \frac{1}{3}\right) = 6 \cdot \frac{1}{6} = 1$, so the numerator is $7 - 1 = 6$.
Step 2: Fraction: $\frac{6}{6} = 1$.
Step 3: $1 + 7(m - 1) = 1 + 7m - 7 = 7m - 6$.

Rabaa Farrag

14 **Answer: C** $AB = BC$

Step 1: A triangle can have at most one obtuse angle, so the base angles are acute and $\angle B$ is the vertex angle.
Step 2: The equal legs of an isosceles triangle meet at the vertex angle.
Step 3: So $AB = BC$.

15 **Answer: B** $45°$

Step 1: In a right triangle the hypotenuse is the longest side; $AB > AC$ means AB is the hypotenuse and the right angle is at C.
Step 2: The legs AC and BC are equal (isosceles), so $\angle A = \angle B$.
Step 3: $\angle A + \angle B = 90° \Rightarrow m\angle B = 45°$.

16 **Answer: C** 5

Step 1: Let $r = 5k$ and $h = 7k$.
Step 2: $V = \pi r^2 h = \pi(25k^2)(7k) = 175\pi k^3 = 550$.
Step 3: $k^3 = \frac{550}{175\pi} \approx 1.0 \Rightarrow k = 1$, so $r = 5$ in.

17 **Answer: D** -4 and 3

Step 1: $|2y + 1| = 7$ means $2y + 1 = 7$ or $2y + 1 = -7$.
Step 2: $2y = 6 \Rightarrow y = 3$; $2y = -8 \Rightarrow y = -4$.
Step 3: Solutions: -4 and 3.

18 **Answer: D** $x + 35 = 65$

Step 1: $\triangle ABC \cong \triangle FED$ gives $\angle A = \angle F = 35°$, $\angle B = \angle E = 30°$, $\angle C = \angle D = 180° - 65° = 115°$.
Step 2: So $x - 45 = 35$, $x - 50 = 30$ and $x + 35 = 115$ all give $x = 80$.
Step 3: $x + 35 = 65$ would give $x = 30$ — it cannot be used.

19 **Answer: B** 157.5

Step 1: Perimeter: $2[(x + 3) + 2x] = 51 \Rightarrow 6x + 6 = 51 \Rightarrow x = 7.5$.
Step 2: Sides: $x + 3 = 10.5$ in and $2x = 15$ in.
Step 3: Area $= 10.5 \times 15 = 157.5$ in^2.

20 **Answer: C** $5\sqrt{6} + 6$

Step 1: Distribute: $\sqrt{6} \cdot 5 + \sqrt{6} \cdot \sqrt{6}$.
Step 2: $= 5\sqrt{6} + 6$.

Rabaa Farrag

21 **Answer: A** 77.5

Step 1: Total of the first 3 exams $= 3 \times 80 = 240$.
Step 2: New total $= 240 + 70 = 310$.
Step 3: New mean $= \frac{310}{4} = 77.5$.

22 **Answer: A** 3

Step 1: Read the bars as frequencies of the values 1 to 7: $1, 5, 3, 2, 1, 4, 1$ — that is 17 data values.
Step 2: The median is the 9th value. Cumulative counts: value $1 \to 1$, value $2 \to 6$, value $3 \to 9$.
Step 3: The 9th value is 3.

23 **Answer: D** $192{,}000$

Step 1: The visitors double every 5 years: $3, 6, 12$ (1970, 1975, 1980).
Step 2: Continue: 24 (1985), 48 (1990), 96 (1995), 192 (2000) thousand.
Step 3: About $192{,}000$ visitors in 2000.

24 **Answer: A** $\overline{PQ} \parallel \overline{TS}$

Step 1: $\angle PRQ \cong \angle SRT$ (vertical angles).
Step 2: If $\overline{PQ} \parallel \overline{TS}$, then $\angle P \cong \angle S$ (alternate interior angles with transversal $\overline{PS}$).
Step 3: Two pairs of equal angles: the triangles are similar by AA.

25 **Answer: C** $55°$

Step 1: $\overline{AD}$ is common, $CD = DB$, and $\angle ADC = \angle ADB = 90°$ (AD is perpendicular to the plane), so $\triangle ADC \cong \triangle ADB$ (SAS).
Step 2: So $m\angle DAC = m\angle DAB = 35°$.
Step 3: In right $\triangle ADC$: $m\angle ACD = 90° - 35° = 55°$.

26 **Answer: A** $x > \frac{c+2b}{a}$

Step 1: Subtract b: $ax > 2b + c$.
Step 2: Divide by a (positive, so the sign stays): $x > \frac{c+2b}{a}$.

27 **Answer: C** -3 and 6

Step 1: $f(2) = 4 + 6 + 2b + 3b = 10 + 5b = -20 \Rightarrow b = -6$.
Step 2: $f(x) = x^2 + 3x - 6x - 18 = x^2 - 3x - 18$.
Step 3: $x^2 - 3x - 18 = (x - 6)(x + 3)$, zeros -3 and 6.

Rabaa Farrag

28 **Answer: C** $12x^4y^8$

Step 1: $x^0 = 1$, so the first factor is $3y^2$.
Step 2: $(2x^2y^3)^2 = 4x^4y^6$.
Step 3: $3y^2 \cdot 4x^4y^6 = 12x^4y^8$.

29 **Answer: C** 15

Step 1: The ratio stays $4:6$, so new height $= 4k$ and new length $= 6k$.
Step 2: Length is 5 more than height: $6k - 4k = 5 \Rightarrow k = 2.5$.
Step 3: New length $= 6(2.5) = 15$ in.

30 **Answer: A** $-\frac{18}{5}$

Step 1: Simplify first: $\frac{4x^5y^2}{15x^2y^3} = \frac{4x^3}{15y}$.
Step 2: Substitute: $\frac{4(27)}{15(-2)} = \frac{108}{-30}$.
Step 3: $= -\frac{18}{5}$.

31 **Answer: B** 284

Step 1: 2 h 20 min $= 2\frac{1}{3} = \frac{7}{3}$ h.
Step 2: Speed $= 662 \div \frac{7}{3} = \frac{662\times3}{7} \approx 283.7$.
Step 3: About 284 mph.

32 **Answer: A** -1 and 4

Step 1: Factor: two numbers with product -4 and sum -3: -4 and 1.
Step 2: $x^2 - 3x - 4 = (x-4)(x+1)$.
Step 3: Zeros: -1 and 4.

33 **Answer: D** $\frac{2\sqrt{3}}{3}$

Step 1: Multiply top and bottom by $\sqrt{3}$: $\frac{2}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$.
Step 2: $= \frac{2\sqrt{3}}{3}$.

34 **Answer: C** $y = x^2 - 9x + 14$

Step 1: The graph is a parabola (two turning directions would mean a cubic), so it is B or C.
Step 2: It passes through $(2, 0)$ and $(3, -4)$. Test C: $2^2 - 9(2) + 14 = 0$ ✓ and $3^2 - 9(3) + 14 = -4$ ✓.
Step 3: (B gives 0 at $x = 3$, not -4.) The equation is $y = x^2 - 9x + 14$.

Rabaa Farrag

35 **Answer: B** Graph B

Step 1: From the table, each time x increases by 1, y decreases by 2.
Step 2: The points lie on a line with slope -2 (steep and decreasing).
Step 3: Graph B is the steep decreasing line.

36 **Answer: D** $\frac{x\sqrt{y}}{y}$

Step 1: Rationalize: multiply by $\frac{\sqrt{y}}{\sqrt{y}}$.
Step 2: $\frac{x}{\sqrt{y}} \cdot \frac{\sqrt{y}}{\sqrt{y}} = \frac{x\sqrt{y}}{y}$.

37 **Answer: A** $F = \frac{9}{5}C + 32$

Step 1: Translate: $9C = 5F - 160$.
Step 2: $5F = 9C + 160 \Rightarrow F = \frac{9C+160}{5}$.
Step 3: $F = \frac{9}{5}C + 32$.

38 **Answer: D** 720

Step 1: One rotation covers one circumference: $2\pi r = 2(3.14)(14) = 87.92$ in.
Step 2: 1 mile $= 5280 \times 12 = 63{,}360$ in.
Step 3: Rotations $= \frac{63360}{87.92} \approx 720.7$, so 720 complete rotations.

39 **Answer: D** $\left\{\frac{3}{2}, \frac{5}{2}\right\}$

Step 1: Factor: $4 \times 15 = 60$; numbers with product 60 and sum -16: -6 and -10.
Step 2: $4x^2 - 6x - 10x + 15 = 2x(2x - 3) - 5(2x - 3) = (2x - 3)(2x - 5)$.
Step 3: $x = \frac{3}{2}$ or $x = \frac{5}{2}$.

40 **Answer: B** 0 lines and 1 point

Step 1: The two semicircles bulge to opposite sides, so a reflection would not map one onto the other: 0 lines of symmetry.
Step 2: Rotating $180°$ about the point where the diameters meet swaps the two semicircles exactly: 1 point of symmetry.

41 **Answer: B** $\frac{5}{2}$

Step 1: Both are right triangles (at B and E) and $\angle C \cong \angle F$, so $\triangle ABC \sim \triangle DEF$ (AA).
Step 2: Corresponding sides: $AB \leftrightarrow DE$ and $BC \leftrightarrow EF$.
Step 3: $\frac{BC}{AB} = \frac{EF}{DE} \Rightarrow \frac{5}{8} = \frac{EF}{4} \Rightarrow EF = \frac{5}{2}$ cm.

Rabaa Farrag

42 **Answer: D** 24

Step 1: $\overline{XY}$ $(25$ in$)$ is the hypotenuse and $XZ = 7$ in is a leg.
Step 2: $ZY = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576}$.
Step 3: $ZY = 24$ in.

43 **Answer: D** $6\sqrt{10}$

Step 1: Multiply the coefficients and the radicands separately: $(2 \cdot 3)\sqrt{2 \cdot 5}$.
Step 2: $= 6\sqrt{10}$.

44 **Answer: C** $8x^4y^6 + 24x^6y^9$

Step 1: $4x^3y^2 \cdot 2y^4x = 8x^4y^6$.
Step 2: $4x^3y^2 \cdot 6x^3y^7 = 24x^6y^9$.
Step 3: Sum: $8x^4y^6 + 24x^6y^9$.

45 **Answer: C** $x > -1$

Step 1: The circle at -1 is open, so -1 itself is not included.
Step 2: The arrow is shaded to the right (greater values).
Step 3: $x > -1$.

46 **Answer: A** Pentagon

Step 1: Interior angle sum: $(n - 2) \cdot 180^\circ = 540^\circ$.
Step 2: $n - 2 = 3 \Rightarrow n = 5$: a pentagon.

47 **Answer: C** 45°

Step 1: $\overline{AC} \perp \overline{BD}$ at the center O, so the central angle $\angle BOC = 90^\circ$ and arc $BC = 90^\circ$.
Step 2: $\angle BEC$ is an inscribed angle that intercepts arc BC.
Step 3: $m\angle BEC = \frac{1}{2}(90^\circ) = 45^\circ$.

48 **Answer: B** 50.4

Step 1: The incline is a right triangle: base 67.2 yd, hypotenuse 84 yd.
Step 2: $h = \sqrt{84^2 - 67.2^2} = \sqrt{7056 - 4515.84} = \sqrt{2540.16}$.
Step 3: $h = 50.4$ yd.

Rabaa Farrag

49 **Answer: B** 24

Step 1: The 48° angle is at the center (a central angle); x is at the circle (an inscribed angle) and both intercept the same arc.
Step 2: Inscribed angle $= \frac{1}{2}$ central angle.
Step 3: $x = \frac{48^\circ}{2} = 24^\circ$.

50 **Answer: B** 77.17

Step 1: The big square (side 24) contains 4 congruent right triangles, each with hypotenuse 24 and a 30° angle.
Step 2: Legs: $24\sin 30^\circ = 12$ and $24\cos 30^\circ = 12\sqrt{3} \approx 20.785$.
Step 3: The shaded square's side is the difference of the legs: $20.785 - 12 = 8.785$. Area $\approx 8.785^2 \approx 77.17$.

Rabaa Farrag